

Phonon-Modulated Hubble Parameter $H(t, \Gamma)$ via F_U and E_{net} Coupling

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Abstract

We derive the phonon-modulated Hubble parameter:

$$H(t, \Gamma) = H_0 \left(1 + \frac{\Phi}{F_U} \cdot E_{\text{net}} \right)$$

This couples the phonon resonance flux $\Phi_{1.25 \text{ THz}}$ to the unified gravitational field F_U and the net vacuum energy E_{net} , producing a dynamical Hubble rate that amplifies during periods of strong phonon-vacuum coupling. Unlike PAPER_1084 (which derives the inflationary H from ρ_{SCm}), this formulation connects H to the present-epoch observables F_U , E_{net} , and Φ .

§1 Core Equation

$$H(t, \Gamma) = H_0 \left(1 + \frac{\Phi(\Gamma)}{F_U(M, r)} \cdot E_{\text{net}}(t) \right)$$

where: - $H_0 = 2.195 \times 10^{-18} \text{ s}^{-1}$ (67.74 km/s/Mpc) - $\Phi(\Gamma) = \Phi_0 \cdot S_{26}$ at resonance - $F_U = \sum_{i=1}^{26} GM/r^2 \cdot [SSq] \cdot i/26$ (26-layer gravity)

§2 Net Vacuum Energy

$$E_{\text{net}}(t) = E_0 \cdot e^{(\kappa + [SSq]/26)t} \cdot S_{26} \cdot (2R - 1)$$

where $R = F_{U, Bi}/F_U$ is the buoyancy ratio: - $R > 0.5$: expansion (positive E_{net}) - $R < 0.5$: erosion (negative E_{net})

§3 Amplification Factor

The phonon amplification of H is:

$$\frac{H(t, \Gamma)}{H_0} = 1 + \frac{\Phi \cdot E_{\text{net}}}{F_U}$$

For solar parameters ($M = M_{\odot}$, $r = 1$ AU, $R = 0.8$): $H/H_0 \sim 10^6$, showing enormous amplification — physically this represents the local departure from Hubble flow near massive objects.

§4 Gamma Dependence

The phonon linewidth Γ enters through $\Phi(\Gamma)$:

$$\Phi(\Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

At resonance: $\Phi = \Phi_0 \cdot S_{26} \approx 1.45 \times 10^{21}$. Off-resonance: Φ decreases Gaussian-sharply with linewidth.

§5 Physical Interpretation

The formula $H(t, \Gamma) = H_0(1 + \Phi/F_U \cdot E_{\text{net}})$ encodes: 1. **Cosmic expansion base rate:** H_0 (Hubble constant) 2. **Local phonon enhancement:** Φ/F_U ratio (SCm vacuum coupling strength) 3. **Net energy direction:** E_{net} sign (expansion vs erosion) 4. **Linewidth modulation:** Γ controls phonon resonance sharpness

References

- PAPER_1084: SCm Phonon-Coupled Inflationary Scale Factor
- PAPER_1073: SCm Phonon-Driven Inflation
- PAPER_897: PhononModulatedEnergyEnetPhononCalc

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